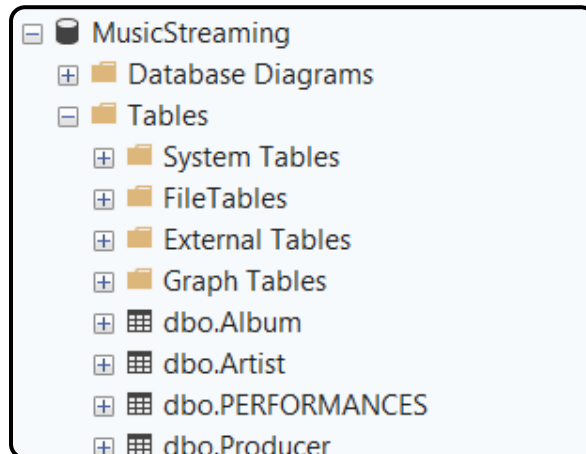


Basic SQL Queries

Remember, SQL is a language for getting information from databases, so *before we code, we need to know what's in our database.*

+ MusicStreaming

+ Tables



At the bottom you see 4 "tables".

- (They're actually **objects not tables**, ".dbo" stands for database object, think of object as the table, so **Album object** is just the **Album Table**)

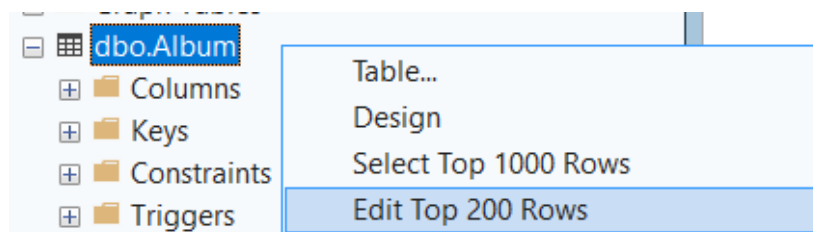
dbo.Album

dbo.Artist

dbo.PERFORMANCES

dbo.Producer

> Right click "dbo.Album" so we can see what we'd actually be "querying".



Without the ability to see what data is in our database, we won't be able to make useful queries.

albumid	albumtitle	releaseyear	genre	tracks	sales	producerid
100	Midnight Rain ...	2020	Pop	13	2500000	4
101	Sunrise Over ...	2021	Pop	12	1800000	4
102	Starlight ...	2022	Pop	15	3200000	4
200	Summer Pop ...	2020	Pop	10	900000	2
201	Dance Hits ...	2021	Pop	8	450000	2
202	Euphoria ...	2023	Pop	14	1300000	2
203	Pop Revolution ...	2022	Pop	12	1100000	2
300	Classic Vibes ...	2021	Rock	10	300000	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL

So, our table (**object**) has these **columns**, these in are called attributes.

- Album's (our table) have an attribute called **albumtitle**.

These **attributes** are what we use to actually do queries with.

Now let's write our **first query**.

The *standard query* is 3 lines.

- **SELECT** (What columns you wish to grab, * for ALL columns from that table)
- **FROM** (Table you wish to grab it from)
- **WHERE** (Relational Filter, <, >, =, ≠)

```
SQLQuery1....Aslam (55))*
1  SELECT albumtitle
2  FROM Album
3  WHERE sales = 450000
```

	albumtitle
1	Dance Hits

In fact, you can also include NOT, OR, AND in your WHERE line.

```
SELECT albumtitle
FROM Album
WHERE sales > 5000 AND sales < 1500000
```

	albumtitle
1	Summer Pop
2	Dance Hits
3	Euphoria
4	Pop Revolution
5	Classic Vibes

You can do a bit with the *last 2 filters* on WHERE alone, but **there are some extra ones you should know about.**

BETWEEN:

- Checks if value *is or is between 2 numbers.*

```
SELECT albumtitle
FROM Album
WHERE sales BETWEEN 1 AND 1000000
```

NULL:

- Checks if value has *anything at all* in it or not.

```
SELECT albumtitle
FROM Album
WHERE sales IS NULL
```

IN:

- Checks if value *matches ANY of the values listed.* If so, then return it

```
SELECT albumtitle
FROM Album
WHERE sales IN (450000, 900000, 1100000)
```

LIKE:

- Only return values that match a **pattern**.
 - **%** and **_** are *placeholder's*. See the [pattern here](#).
 - **%night%** → Anything that has "night" in it, **anything after or before**.
 - **night%** → Anything that **STARTS** with night, **anything after**.
 - **%night** → Anything that **ENDS** with night, **anything before**.
 - **night_** → "night" but has *room for one* random character **after it**.
 - **_night** → "night" but has *room for one* random character **before it**.

Looks like this →

```
SELECT albumtitle
FROM Album
WHERE albumtitle LIKE '%night%'
```

	albumtitle
1	Midnight Rain

Those are the **BASICS**, for querying in SQL. Next lab we'll cover more Math related filters for getting information.